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## Exp 6 :Production Rules and Forward Chaining

### 1. Aim

To implement production rules using forward chaining in Python to derive new facts from known facts.

### 2. Algorithm

1. Start the program.
2. Define initial facts and IF–THEN rules.
3. Check rule conditions against known facts.
4. Add new conclusions when conditions match.
5. Display the final facts and stop.

**Program:**

```
facts = {"fever", "cough"}


rules = [
   ({"fever"}, "possible_illness"),
   ({"fever", "cough"}, "possible_infection")
]


changed = True


while changed:
   changed = False
   for conditions, conclusion in rules:
      if conditions.issubset(facts) and conclusion not in facts:
         facts.add(conclusion)
         changed = True
```

```
print("Final Facts:")
for fact in sorted(facts):
    print(fact)
```

**INPUT:**

fever

cough

**OUTPUT:**

```
========== RESTART: C:/Users/Divya/Downloads/knowledge lab/exp6.py ==========
Final Facts:
cough
fever
possible_illness
possible_infection
```

**RESULT:**

Production rules and forward chaining were implemented successfully using Python.